

Homework 1

Due Wednesday, August 30

1. The length of a curve on a plane in Cartesian coordinates, parametrized by $x(t)$, $y(t)$, is

$$l = \int_{t_1}^{t_2} [\dot{x}^2 + \dot{y}^2]^{1/2} dt.$$

- (a) Obtain the corresponding expression in polar coordinates $r(t)$, $\phi(t)$.
 - (b) Using that expression, obtain the Euler-Lagrange (EL) equations for $r(t)$, $\phi(t)$.
 - (c) Excluding the “time” t , solve the EL equations to obtain $\phi(r)$. The answer should contain two integration constants, i.e., describe a two-parametric family (of geodesic curves on the plane).
2. The action of a particle of mass m moving in a 3-dimensional anti-de Sitter (AdS₃) spacetime is

$$S = -m \int dt \left[e^{-2kz} (1 - \dot{x}^2) - \dot{z}^2 \right]^{1/2},$$

where m is the mass of the particle, and $k > 0$ is a constant. There are two generalized coordinates, x and z (although time t has been included in the counting of dimensions). We use the system of units where the speed of light is $c = 1$.

- (a) Obtain the canonical momenta conjugate to x and z and the Hamiltonian H .
- (b) Because the Lagrangian does not explicitly depend on time, the Hamiltonian is also a conserved energy, i.e., $H = E$ is the first integral. Suppose the particle starts at time $t = 0$ at $z = 0$ and arbitrary x with zero velocity. Find $x(t)$, $z(t)$ for all future times. *Note:* As a check of your calculation, you may observe that it takes infinite time for the particle to reach $z = \infty$. On the other hand, the **proper** time required for that is finite.